Cascaded $\chi^{(2)}{:}\chi^{(2)} \to \chi^{(3)}_{\text{eff}}$ in a Singly-Resonant (SH-only) Metasurface

derivation and comparison with the doubly-resonant case

October 6, 2026

1 Setting and goal

The companion note *Cascaded $\chi^{(3)}$ in a doubly-resonant dielectric metasurface* derives the cascaded Kerr response when both the fundamental (ω_1) and the second harmonic ($\omega_2 \approx 2\omega_1$) are sharp resonances of the same meta-atom. Here we ask what changes when the metasurface has *only* the SH resonance: the FM is in the linear transmission band, with no field build-up at ω_1.

This is a relevant question for two reasons. First, designing two spectrally-aligned co-localised resonances (qBIC at ω_1 and a high-Q BIC near $2\omega_1$) requires significant mode-engineering and fabrication control; a single SH resonance is much cheaper. Second, the SH-only configuration is the cleanest experimental platform for *isolating* the $\chi^{(2)}{\cdot}G{\cdot}\chi^{(2)}$ kernel from FM-lineshape effects, because the FM transmission is featureless across the relevant tuning range. We therefore want the cascaded $\chi^{(3)}$ for this geometry in closed form, with the same conventions and overlap structure as in the main note, so the two boxed results can be compared term by term.

Result preview. The structural form $\chi^{(3)}_{\text{casc}} \simeq -2\omega_1\,|\chi^{(2)}|^2\,\eta_{\text{geom}}/(3\,n_2^2\,\Delta\omega)$ *survives* the loss of the FM resonance. The geometric overlap η_{geom} takes a different (typically smaller) value, set by how well the spatially-uniform incident pump can drive $\mathbf{f}_2$ through $\chi^{(2)}(\mathbf{r})$. The headline change is *not* in $\chi^{(3)}_{\text{casc}}$ itself but in the per-photon nonlinear phase observable on a finite-thickness metasurface, which loses the Q_1^2 build-up factor that doubly-resonant operation provides. Section 7 makes this quantitative.

Conventions. Boyd field convention $\mathcal{E} = Ee^{-i\omega t} + \text{c.c.}$ (no prefactor $\frac{1}{2}$), SHG and DFG polarisations with degeneracies 1 and 2 respectively, SPM with degeneracy 3, Kleinman symmetry, Brillouin energy normalisation $\int[\varepsilon_0\widetilde{\varepsilon}_n|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2]\,d^3r = 1$, so that $|a_n|^2$ is the stored energy and the projection norm $N_n \equiv \frac{1}{2}\int[\varepsilon_0\widetilde{\varepsilon}_n|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2] = \frac{1}{2}$. All identical to the main note's §2–§3 (as corrected in October 2026: an earlier version carried a factor-4 error in the cascaded prefactor, which this note inherited; see the main note, §5). Frequency detuning $\Delta\omega \equiv 2\omega_1 - \omega_2$.

2 The FM field without a resonance

In the doubly-resonant case the FM field inside the meta-atom is $\mathbf{E}(\mathbf{r},\omega_1) = a_1\,\mathbf{f}_1(\mathbf{r})$, with the modal amplitude $a_1 \in \mathbb{C}$ carrying the resonant build-up: $|a_1|^2 \sim |s_+|^2\,Q_1/\omega_1$ at coupled-FM resonance.

With no FM resonance, the modal expansion of $\mathbf{E}(\omega_1)$ has no large near-pole term. In the linear (zero-nonlinearity) limit at ω_1, write the field inside the meta-atom as

$$\mathbf{E}(\mathbf{r},\omega_1) \;=\; \mathbf{E}_{\rm loc}(\mathbf{r})\,, \tag{1}$$

the steady-state response of the metasurface to an incident plane wave $\mathbf{E}_{\rm in}(\mathbf{r},\omega_1) = E_0\,\hat{\boldsymbol{e}}\,e^{i\mathbf{k}_1\cdot\mathbf{r}}$ at frequency ω_1, computed in the absence of nonlinearity. $\mathbf{E}_{\rm loc}(\mathbf{r})$ is a c-number field, not a dynamical mode amplitude: it is fixed by the incident drive and does not have an independent equation of motion.

Magnitude and spatial profile. Inside a high-index dielectric meta-atom of refractive index n_1 embedded in air, the off-resonant Mie / slab-Fresnel regime gives

$$|\mathbf{E}_{\rm loc}(\mathbf{r})| \;\sim\; \frac{T_{\rm geom}}{n_1}\,E_0, \qquad \text{(inside the dielectric, for } \mathbf{r}\in V_{\rm NL}\text{)}, \tag{2}$$

with $T_{\rm geom}\sim\mathcal{O}(1)$ a geometric prefactor absorbing the shape-dependent depolarisation factor and Fresnel transmission at the interface. (Two limiting cases bracket this: a thin slab gives $2/(1+n_1)\sim 1/n_1$, and a small dielectric sphere in the Clausius–Mossotti limit gives $3/(n_1^2+2)\sim 1/n_1^2$ – typical sub-wavelength $LiNbO_3$ pillars sit closer to the slab/Fresnel regime because the depth of the meta-atom is not deep in the quasi-static limit; we adopt $1/n_1$ as the conservative default and absorb $\mathcal{O}(1)$ shape residuals into $T_{\rm geom}$.) The crucial point is the absence of any factor of Q_1: the linear FM response is broadband, and a plane-wave intensity $I_{\rm in} = 2\varepsilon_0 c n_{\rm ext}|E_0|^2$ (Boyd's convention; $n_{\rm ext}=1$ in air) translates to local electric energy density $\varepsilon_0 n_1^2|\mathbf{E}_{\rm loc}|^2 \sim T_{\rm geom}^2\,I_{\rm in}/(2c)$ inside the dielectric, with no resonant enhancement.

Polarisation pattern. For a meta-atom whose lowest electric multipole is dipolar, $\mathbf{E}_{\rm loc}(\mathbf{r})$ is to leading order spatially uniform inside $V_{\rm NL}$ and parallel to $\hat{\boldsymbol{e}}$. We will use this uniform-pump approximation throughout, so

$$\mathbf{E}_{\rm loc}(\mathbf{r}) \;\approx\; E_{\rm loc}\,\hat{\boldsymbol{e}}, \qquad E_{\rm loc}\equiv\frac{T_{\rm geom}}{n_1}\,E_0, \qquad \mathbf{r}\in V_{\rm NL}. \tag{3}$$

Corrections to this approximation are at most $\mathcal{O}(1)$ shape factors inside the meta-atom and can be absorbed into a redefinition of $T_{\rm geom}$; they do not alter the structural form derived below.

3 TCMT for the SH-only mode

Repeating the projection-onto-$\mathbf{f}_2^*$ argument of the main note's §5 with the SHG polarisation source $P_i^{(2)}(\mathbf{r},2\omega_1) = \varepsilon_0\,\chi_{ijk}^{(2)}(\mathbf{r})\,E_{{\rm loc},j}(\mathbf{r})\,E_{{\rm loc},k}(\mathbf{r})$ gives the SH-mode equation

$$\boxed{\frac{da_2}{dt} = \left(i\Delta\omega - \tfrac{\gamma_2}{2}\right)a_2 \;+\; i\,\hat{\beta}\,,} \tag{4}$$

where the SH-mode amplitude a_2 is again Brillouin-normalised so that $|a_2|^2$ is the stored SH energy. The crucial difference from the doubly-resonant equation $\dot{a}_2 = (i\Delta\omega-\gamma_2/2)a_2 + i\beta a_1^2$ is that the drive $\hat{\beta}$ is a c-number set by the incident pump – *not* a quadratic function of a dynamical FM amplitude. Explicitly,

$$\boxed{\hat{\beta} \;=\; \frac{\varepsilon_0\,\omega_1}{N_2}\int_{V_{\rm NL}}\chi_{ijk}^{(2)}(\mathbf{r})\,f_{2,i}^*(\mathbf{r})\,E_{{\rm loc},j}(\mathbf{r})\,E_{{\rm loc},k}(\mathbf{r})\;d^3r\,,} \qquad [\text{units}:\ \mathrm{J}^{1/2}\mathrm{s}^{-1}]. \tag{5}$$

This is the singly-resonant analogue of the main note's coupling $\beta = (\varepsilon_0\omega_1/N_2)\int \chi^{(2)} f_2^* f_1 f_1 \, d^3r$ (main note Eq. (beta)); with $N_2 = \frac{1}{2}$ the prefactor is $2\varepsilon_0\omega_1$. It follows from the same projection of Maxwell's equations onto $\mathbf{f}_2^*$ (main note Eq. (projection)), so the $\hat{\beta} \leftrightarrow \beta$ correspondence is exact at the prefactor level. (An earlier version used $\varepsilon_0\omega_1/4$ at $N_2 = 1$, which is $4\times$ too small.)

Steady-state SH amplitude. Equation (4) is driven at frequency $2\omega_1$ (in the lab frame) by the externally-imposed $\hat{\beta}$. Setting $\dot{a}_2 \to 0$,

$$a_2^{\mathrm{ss}} \;=\; \frac{i\,\hat{\beta}}{\gamma_2/2 - i\,\Delta\omega} \;=\; -\,\frac{\hat{\beta}}{\Delta\omega + i\,\gamma_2/2}\,, \tag{6}$$

identical in structure to the doubly-resonant $a_2 = -\beta a_1^2/(\Delta\omega + i\gamma_2/2)$ with the replacement $\beta\, a_1^2 \;\longleftrightarrow\; \hat{\beta}$. In the doubly-resonant case $|\beta a_1^2|$ scales with the FM-resonant build-up $|a_1|^2 \propto Q_1$; here $|\hat{\beta}| \propto |E_{\mathrm{loc}}|^2 \propto |E_0|^2/n_1^2$ has no such enhancement. Quantifying this gap is the main work of Section 7.

No FM equation. There is no a_1 degree of freedom and hence no equation for it. The DFG back-action at ω_1 does not feed an FM resonance; instead it generates a local nonlinear polarisation $P^{\mathrm{casc}}(\mathbf{r}, \omega_1)$ that radiates and modifies the transmitted FM field directly through the metasurface response. We derive this polarisation next.

4 Cascaded back-polarisation at ω_1

The SH field generated inside the meta-atom is, in the single-resonance approximation,

$$E_m(\mathbf{r}, 2\omega_1) \;=\; a_2^{\mathrm{ss}}\, f_{2,m}(\mathbf{r}) \;=\; -\,\frac{\hat{\beta}\, f_{2,m}(\mathbf{r})}{\Delta\omega + i\,\gamma_2/2}\,. \tag{7}$$

Substituting into the DFG polarisation source $P_i^{(2)}(\mathbf{r}, \omega_1) = 2\varepsilon_0 \chi_{ikm}^{(2)}(\mathbf{r})\, E_m(\mathbf{r}, 2\omega_1) E_k^*(\mathbf{r}, \omega_1)$ and using $E_k^*(\mathbf{r}, \omega_1) = E_{\mathrm{loc},k}^*(\mathbf{r})$ from (1),

$$P_i^{\mathrm{casc}}(\mathbf{r}, \omega_1) \;=\; -\,\frac{2\varepsilon_0\,\hat{\beta}}{\Delta\omega + i\,\gamma_2/2}\, \chi_{ikm}^{(2)}(\mathbf{r})\, f_{2,m}(\mathbf{r})\, E_{\mathrm{loc},k}^*(\mathbf{r})\,. \tag{8}$$

Two structural features deserve emphasis.

Non-locality reduces to a global amplitude $\times$ local pattern. Equation (8) is the singly-resonant collapse of the general $\chi^{(2)}{\cdot}G(2\omega_1){\cdot}\chi^{(2)}$ kernel of the main note's §4 (the boxed non-local $\chi^{(2)}{\cdot}G{\cdot}\chi^{(2)}$ kernel). The non-local integral over $\mathbf{r}'$ has been performed exactly by the single-pole QNM expansion of $G(2\omega_1)$: the SH-driving overlap $\hat{\beta}$ in (5) integrates the $\chi^{(2)}{:}\mathbf{E}_{\mathrm{loc}}\mathbf{E}_{\mathrm{loc}}$ source against $\mathbf{f}_2^*$ at $\mathbf{r}'$, while the remaining local factor $\chi_{ikm}^{(2)}(\mathbf{r}) f_{2,m}(\mathbf{r}) E_{\mathrm{loc},k}^*(\mathbf{r})$ at $\mathbf{r}$ has the spatial pattern of the SH mode profile, weighted by $\chi^{(2)}(\mathbf{r})$. In the doubly-resonant case the local factor projects further onto $\mathbf{f}_1^*$ to yield an SPM-shaped Kerr; here, in the absence of $\mathbf{f}_1$, the cascaded polarisation *retains* its $\chi^{(2)} \cdot \mathbf{f}_2$ spatial signature.

Kerr-like cubic dependence on the pump. $\hat{\beta} \propto E_{\mathrm{loc}}^2$ and the second power of $\mathbf{E}_{\mathrm{loc}}$ in (5), combined with the explicit $E_{\mathrm{loc},k}^*$ in (8), give a polarisation that is bilinear in $\mathbf{E}_{\mathrm{loc}}^*{\cdot}\mathbf{E}_{\mathrm{loc}}$ and linear in $\mathbf{E}_{\mathrm{loc}}$, i.e. the standard $|E|^2E$ structure of an SPM-type Kerr. This is what justifies calling (8) an "effective $\chi^{(3)}$" response at ω_1 even without an FM modal description.

Substituting $\hat{\beta}$ explicitly. Using (5) in (8),

$$P_i^{\mathrm{casc}}(\mathbf{r},\omega_1) \;=\; -\,\frac{2\varepsilon_0^2\,\omega_1/N_2}{\Delta\omega + i\,\gamma_2/2}\;\chi^{(2)}_{ikm}(\mathbf{r})\,f_{2,m}(\mathbf{r})\,E^{*}_{\mathrm{loc},k}(\mathbf{r}) \times \int_{V_{\mathrm{NL}}} \chi^{(2)}_{jpq}(\mathbf{r}')\,f^{*}_{2,j}(\mathbf{r}')\,E_{\mathrm{loc},p}(\mathbf{r}')\,E_{\mathrm{loc},q}(\mathbf{r}')\,d^3r'. \quad (9)$$

Under the uniform-pump approximation (3) (field $E_{\mathrm{loc}}\,\hat{\boldsymbol{e}}$ inside the dielectric), define the two scalar overlaps

$$\widetilde{M} \equiv \int_{V_{\mathrm{NL}}} \chi^{(2)}_{jpq}(\mathbf{r}')\,f^{*}_{2,j}(\mathbf{r}')\,\hat{e}_p\,\hat{e}_q\,d^3r', \quad (10)$$

$$\widetilde{M}'_i \equiv \int_{V_{\mathrm{NL}}} \chi^{(2)}_{ikm}(\mathbf{r})\,f_{2,m}(\mathbf{r})\,\hat{e}^{*}_k\,d^3r. \quad (11)$$

By Kleinman symmetry $\chi^{(2)}_{ikm} = \chi^{(2)}_{mki} = \chi^{(2)}_{kmi} = \cdots$, and for a real pump polarisation $\hat{\boldsymbol{e}}^{*} = \hat{\boldsymbol{e}}$, $\widetilde{M}'_i\,\hat{e}_i = \widetilde{M}^{*}$ (relabelling indices and using full permutation symmetry of the three pairs in $\chi^{(2)}_{jpq}$ followed by complex conjugation of the SH-mode profile). Carrying out the $\mathbf{r}'$ integration in (9) produces *one* factor of $\widetilde{M}$ multiplying the local pattern at $\mathbf{r}$:

$$P_i^{\mathrm{casc}}(\mathbf{r},\omega_1)\,\hat{e}_i \;=\; -\,\frac{2\varepsilon_0^2\,\omega_1/N_2}{\Delta\omega + i\,\gamma_2/2}\;\widetilde{M}\;|E_{\mathrm{loc}}|^2\,E_{\mathrm{loc}}\,\big[\chi^{(2)}_{ikm}(\mathbf{r})\,f_{2,m}(\mathbf{r})\,\hat{e}_k\,\hat{e}_i\big], \quad (12)$$

the bracketed factor being r-dependent. The second power of $\widetilde{M}$ will appear only after the $\mathbf{r}$ integration in Section 5, where the bracketed factor integrates to $\widetilde{M}'_i\,\hat{e}_i = \widetilde{M}^{*}$ by Kleinman.

5 Effective $\chi^{(3)}_{\mathbf{casc},(1\mathrm{R})}$ of the SH-only metasurface

A spatially-resolved $\chi^{(3)}(\mathbf{r})$ is not the right object for a sub-wavelength metasurface: the relevant observable is the unit-cell dipole moment, which determines the radiated field at ω_1. We therefore define $\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}$ such that a uniform slab of this susceptibility filling the unit cell would radiate the same $|E|^2E$-dependent far field.

Unit-cell-averaged cascaded polarisation. Integrate (9) over the unit cell, project onto $\hat{\boldsymbol{e}}$:

$$\int_{V_{\mathrm{cell}}} P_i^{\mathrm{casc}}(\mathbf{r},\omega_1)\,\hat{e}_i\;d^3r \;=\; -\,\frac{2\varepsilon_0^2\,\omega_1/N_2}{\Delta\omega + i\,\gamma_2/2}\;|\widetilde{M}|^2\,|E_{\mathrm{loc}}|^2\,E_{\mathrm{loc}}\,, \quad (13)$$

where we have used (11) to perform the $\mathbf{r}$ integration (the integrand vanishes outside V_{NL} by construction). Equating to the dipole moment of a fictitious uniform-slab Kerr $\overline{P}^{(3)}_{\mathrm{eff}} = 3\varepsilon_0\,\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}\,|E_{\mathrm{loc}}|^2E_{\mathrm{loc}}$ filling the same unit cell,

$$3\,\varepsilon_0\,\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}\,|E_{\mathrm{loc}}|^2E_{\mathrm{loc}}\,V_{\mathrm{cell}} \;=\; -\,\frac{2\varepsilon_0^2\,\omega_1/N_2}{\Delta\omega + i\,\gamma_2/2}\;|\widetilde{M}|^2\,|E_{\mathrm{loc}}|^2E_{\mathrm{loc}}\,, \quad (14)$$

and dividing through,

$$\boxed{\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}(\omega_1) \;=\; -\,\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2}\;\frac{|\widetilde{M}|^2}{(\Delta\omega + i\,\gamma_2/2)\,V_{\mathrm{cell}}}\,.} \quad (15)$$

This is the singly-resonant analogue of the doubly-resonant boxed result of the main note Eq. (chi3casc),

$$\chi^{(3)}_{\mathrm{casc},(2\mathrm{R})} \;=\; -\,\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2}\,\frac{|M|^2}{(\Delta\omega + i\gamma_2/2)\int|\mathbf{f}_1|^4\,d^3r}\,.$$

The two formulas have *exactly the same structure*; only the overlap and the normalisation volume differ:

$$\frac{|M|^2}{\int|\mathbf{f}_1|^4\,d^3r} \;\longleftrightarrow\; \frac{|\widetilde{M}|^2}{V_{\mathrm{cell}}}. \tag{16}$$

Scaling out canonical units. Using $|\mathbf{f}_2|^2 \sim 1/(2N_2\,\varepsilon_0\,n_2^2\,V_{\mathrm{eff}})$ from the Brillouin normalisation, $|\widetilde{M}|^2/N_2 \sim |\chi^{(2)}|^2 V_{\mathrm{NL}}^2/(2N_2^2\varepsilon_0 n_2^2 V_{\mathrm{eff}})$; the combination $|\widetilde{M}|^2/N_2$ is independent of how $\mathbf{f}_2$ is normalised. Define the singly-resonant geometric overlap

$$\eta^{(1\mathrm{R})}_{\mathrm{geom}} \;\equiv\; \frac{|\widetilde{M}|^2}{N_2\,V_{\mathrm{cell}}}\cdot\frac{\varepsilon_0\,n_2^2}{|\chi^{(2)}|^2} \;\sim\; \frac{V_{\mathrm{NL}}^2}{V_{\mathrm{eff}}\,V_{\mathrm{cell}}}\,, \tag{17}$$

which is $\mathcal{O}(1)$ in the optimistic limit $V_{\mathrm{eff}} = V_{\mathrm{NL}} = V_{\mathrm{cell}}$ (mode tightly confined to a fully nonlinear meta-atom that fills the unit cell), and reduces by the partial-filling factor $V_{\mathrm{NL}}/V_{\mathrm{cell}}$ when the nonlinear region is smaller than the unit cell. With this definition,

$$\boxed{\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}(\omega_1) \;\simeq\; -\,\frac{2\,\omega_1}{3\,n_2^2\,\Delta\omega}\,|\chi^{(2)}|^2\,\eta^{(1\mathrm{R})}_{\mathrm{geom}}\,,} \tag{18}$$

identical in form to the doubly-resonant $\chi^{(3)}_{\mathrm{casc},(2\mathrm{R})} \simeq -2\omega_1|\chi^{(2)}|^2\eta_{\mathrm{geom}}/(3n_2^2\Delta\omega)$, with the geometric overlap reinterpreted. (The value of $\eta^{(1\mathrm{R})}_{\mathrm{geom}}$ for given fields is unchanged by the October 2026 correction; only the prefactor in front of it moved from 1/6 to 2/3.)

6 What changes in η_{geom}

The doubly-resonant geometric overlap is $\eta_{\mathrm{geom}} = \widetilde{\eta}^2/\zeta_1$ with

$$\widetilde{\eta}^2 \propto \left|\int\chi^{(2)}_{ijk}(\mathbf{r})\,f^*_{2,i}\,f_{1,j}\,f_{1,k}\;d^3r\right|^2, \tag{19}$$

$$\zeta_1 \propto \int|\mathbf{f}_1|^4\,d^3r\,. \tag{20}$$

In the singly-resonant case, with the pump uniform inside the meta-atom and of fixed polarisation $\hat{\boldsymbol{e}}$, the integrand $f_{1,j}f_{1,k}$ is replaced by $\hat{e}_j\hat{e}_k$; the normalisation by $\int|\mathbf{f}_1|^4$ is replaced by the unit-cell volume V_{cell}.

Symmetry consequence (load-bearing). $\mathbf{f}_1(\mathbf{r})$ in the 2R case is engineered to be a qBIC – a mode that couples to free space precisely because of a small symmetry-breaking perturbation. Its squared pattern $f_{1,j}(\mathbf{r})f_{1,k}(\mathbf{r})$ can be deliberately matched to the parity of $\mathbf{f}_2(\mathbf{r})$ (symmetry-protected BIC near $2\omega_1$), so that $\widetilde{\eta}^2$ is not symmetry-forbidden even when $\mathbf{f}_2$ has odd parity under a unit-cell symmetry.

Replacing $f_{1,j}f_{1,k} \to \hat{e}_j\hat{e}_k$ removes one major spatial degree of freedom from the integrand. If $\mathbf{f}_2$ has odd parity under, e.g., $\sigma_x : x \to -x$ (typical of a symmetry-protected BIC), then for *unpatterned* $\chi^{(2)}$ (uniform $\chi^{(2)}_{jpq}$ inside V_{NL}) the constant tensor pulls out of (10), $\widetilde{M} = \big(\chi^{(2)}_{jpq}\hat{e}_p\hat{e}_q\big)\int_{V_{\mathrm{NL}}} f^*_{2,j}\,d^3r = 0$ by symmetry of $\mathbf{f}_2$ over the unit cell. Three escape routes restore a nonzero $\widetilde{M}$:

- **Patterned $\chi^{(2)}(\mathbf{r})$.** Periodically-poled $\chi^{(2)}(\mathbf{r})$ with a real-space periodicity matched to the spatial Fourier content of $\mathbf{f}_2(\mathbf{r})$ provides the odd symmetry needed to make $\widetilde{M} \neq 0$. This is the analogue of quasi-phase-matching, applied at the unit-cell scale rather than over millimetres. For $LiNbO_3$ meta-atoms with sub-domain poling, $\eta_{\text{geom}}^{(1\text{R})}$ can reach $\sim 0.05{-}0.2$ – typically a factor $\sim$ few below the doubly-resonant value.
- **Quasi-BIC at $2\omega_1$.** A symmetry-broken $\mathbf{f}_2$ (qBIC instead of BIC) acquires a non-vanishing spatial average, with magnitude controlled by the asymmetry parameter α as $\bar{\mathbf{f}}_2 \propto \alpha$ and $\gamma_2^{(\text{rad})} \propto \alpha^2$ [1]. The relevant figure of merit is $|\widetilde{M}|^2 \cdot Q_2$, with two regimes:
  - *Radiation-limited Q* ($\gamma_2^{(\text{rad})} \gg \gamma_2^{(\text{nr})}$): $Q_2 \propto 1/\alpha^2$ tracks the radiative channel, so $|\widetilde{M}|^2 Q_2 \propto \alpha^2/\alpha^2 = \mathcal{O}(1)$ – a flat trade-off contour, with $\eta_{\text{geom}}^{(1\text{R})}$ and Q_2 exchanged one-for-one.
  - *Absorption-limited Q* ($\gamma_2^{(\text{nr})} \gtrsim \gamma_2^{(\text{rad})}$): Q_2 is set by intrinsic absorption and is α-independent up to the value of α at which $\gamma_2^{(\text{rad})}$ catches up with $\gamma_2^{(\text{nr})}$. In this regime $|\widetilde{M}|^2 Q_2 \propto \alpha^2$ and one wins directly by increasing α. This is the main-note design target for the SH mode (under-coupled qBIC dominated by intrinsic absorption), so the qBIC route is the most productive of the three escape paths.
- **Oblique incidence.** Tilting the pump introduces a nonzero in-plane wave vector $\mathbf{k}_\parallel$ that breaks the parity used in symmetry-protecting the BIC; the SH mode then couples to oblique plane waves and acquires a non-vanishing $\widetilde{M}$. Bandwidth and angle of incidence become the new design knobs.

In all three escape routes the singly-resonant $\eta_{\text{geom}}^{(1\text{R})}$ is typically a factor $\sim 2{-}10$ below the doubly-resonant η_{geom} for comparable meta-atom material choice. We will quote $\eta_{\text{geom}}^{(1\text{R})} \sim 0.1$ as a working number, against the main note's $\eta_{\text{geom}} \sim 0.3$.

7 Comparison with the doubly-resonant cascade

The comparison has two layers. At the level of the effective material susceptibility $\chi_{\text{casc}}^{(3)}$ (in m^2/V^2), the two cases differ only by $\eta_{\text{geom}}^{(1\text{R})}/\eta_{\text{geom}}$, a factor of order $1/3{-}1/10$. At the level of the observable per-photon nonlinear phase φ_{NL}, the gap is much larger, typically $10^5{-}10^7$, because the doubly-resonant case *also* field-builds the FM by Q_1.

Layer 1: $\chi_{\text{casc}}^{(3)}$ ratio

Comparing (18) with the doubly-resonant boxed result,

$$\frac{\chi_{\text{casc},(1\text{R})}^{(3)}}{\chi_{\text{casc},(2\text{R})}^{(3)}} = \frac{\eta_{\text{geom}}^{(1\text{R})}}{\eta_{\text{geom}}}, \tag{21}$$

at fixed $\Delta\omega$, Q_2, and material. Taking the working values $\eta_{\text{geom}} \sim 0.3$ and $\eta_{\text{geom}}^{(1\text{R})} \sim 0.1$, the singly-resonant susceptibility is suppressed by only a factor of ~ 3 relative to the doubly-resonant value. For the $LiNbO_3$ metasurface at $Q_2 = 4\times10^3$, $\Delta\omega = 2\gamma_2$, $\eta_{\text{geom}}^{(1\text{R})} = 0.1$, $|\chi^{(2)}| =$

$5\times10^{-11}\,\mathrm{m/V}$, $n_2 = 2.26$:

$$|\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}| \;\simeq\; \frac{\omega_1\,|\chi^{(2)}|^2\,\eta^{(1\mathrm{R})}_{\mathrm{geom}}}{3\,n_2^2\,\gamma_2} \;=\; \frac{Q_2\,|\chi^{(2)}|^2\,\eta^{(1\mathrm{R})}_{\mathrm{geom}}}{6\,n_2^2}$$
$$\sim\; \frac{4\times10^3\,(5\times10^{-11})^2\,0.1}{6\,(2.26)^2} \;\sim\; 3\times10^{-20}\,\mathrm{m^2/V^2}\,, \tag{22}$$

i.e. about 16× the native $LiNbO_3$ Kerr $\chi^{(3)} \sim 2\times10^{-21}\,\mathrm{m^2/V^2}$. A real enhancement at the *material* level. The doubly-resonant case at the same conditions gives $|\chi^{(3)}_{\mathrm{casc},(2\mathrm{R})}| \sim 1\times10^{-19}\,\mathrm{m^2/V^2}$ (main note §10), a factor ~ 3 larger as expected from (21).

Layer 2: per-photon nonlinear phase

The observable that the deck and main note quote is the FM nonlinear phase φ_{NL} per incident photon. This is where the two cases diverge dramatically.

Doubly-resonant. Main note §11 gives, in the lossless / fully-radiative ($\gamma_1^{(\mathrm{rad})} = \gamma_1$) limit,

$$\varphi^{(2\mathrm{R})}_{\mathrm{NL}} \;=\; \frac{8\,\alpha_3^{\mathrm{eff}}\,|s_+|^2}{\gamma_1^2} \;=\; -\,\frac{8\,|\beta|^2\,|s_+|^2}{\gamma_1^2\,\Delta\omega}\,, \tag{23}$$

with $|s_+|^2$ the incident power. The factor $\gamma_1^{-2} \sim Q_1^2/\omega_1^2$ is the FM field-buildup squared: $|a_1|^2 \propto |s_+|^2/\gamma_1$, and the cubic SPM produces $\varphi \propto |a_1|^2 \propto Q_1$, while a second $1/\gamma_1$ comes from the phase being read out over the cavity lifetime.

Singly-resonant. With no FM resonance the phase shift is acquired in a single pass through the metasurface of thickness $L_{\mathrm{meta}} \sim \lambda_1/(2n_1) - \lambda_1$ (sub-wavelength):

$$\varphi^{(1\mathrm{R})}_{\mathrm{NL}} \;=\; \frac{\omega_1}{c}\,n_1\,\frac{3\,\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}\,|E_{\mathrm{loc}}|^2}{2\,n_1^2}\,L_{\mathrm{meta}} \;=\; \frac{3\,\omega_1}{2\,c\,n_1}\,\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}\,|E_{\mathrm{loc}}|^2\,L_{\mathrm{meta}}, \tag{24}$$

using the SVEA relation $d\varphi/dz = (\omega_1/(2n_1c))\cdot 3\,\chi^{(3)}|E|^2$. Substituting

$$|E_{\mathrm{loc}}|^2 \;=\; \frac{T^2_{\mathrm{geom}}\,P_{\mathrm{in}}}{2\,\varepsilon_0\,c\,n_1^2\,A_{\mathrm{cell}}}$$

from (2), with $|s_+|^2 = P_{\mathrm{in}}$ the incident power per unit cell,

$$\varphi^{(1\mathrm{R})}_{\mathrm{NL}} \;\sim\; \frac{3\,\omega_1\,L_{\mathrm{meta}}\,T^2_{\mathrm{geom}}}{4\,\varepsilon_0\,c^2\,n_1^3\,A_{\mathrm{cell}}}\,\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}\,|s_+|^2\,. \tag{25}$$

The ratio. Forming $\varphi^{(2\mathrm{R})}_{\mathrm{NL}}/\varphi^{(1\mathrm{R})}_{\mathrm{NL}}$ at the same incident power $|s_+|^2$ and the same $\Delta\omega$, and using the swap rule (16) together with $|\beta|^2 \propto |\chi^{(2)}|^2\eta_{\mathrm{geom}}/(\varepsilon_0 n_2^2)$ in the 2R case,

$$\boxed{\frac{\varphi^{(2\mathrm{R})}_{\mathrm{NL}}}{\varphi^{(1\mathrm{R})}_{\mathrm{NL}}} \;\sim\; \underbrace{\frac{\eta_{\mathrm{geom}}}{\eta^{(1\mathrm{R})}_{\mathrm{geom}}}}_{\sim 3} \times \underbrace{\left(\frac{Q_1}{\omega_1 L_{\mathrm{meta}}/c}\right)^2}_{\sim Q_1^2\cdot(\lambda_1/L_{\mathrm{meta}})^2}\,.} \tag{26}$$

For $Q_1 = 10^3$ (a modest qBIC), $\lambda_1 = 1.55\,\mu\mathrm{m}$, $L_{\mathrm{meta}} = 400\,\mathrm{nm}$, so $\omega_1 L_{\mathrm{meta}}/c = 2\pi L_{\mathrm{meta}}/\lambda_1 \simeq 1.62$ and the second factor is $(Q_1/1.62)^2 \simeq 3.8\times10^5$; with the geometry factor ~ 3 in front, and the $\mathcal{O}(1)$ prefactor $8\zeta_1 V_{\mathrm{cell}}/(n_1 T^2_{\mathrm{geom}} V_{\mathrm{eff}}) \approx 3.6$ that the scaling form (26) omits ($V_{\mathrm{eff}} \approx V_{\mathrm{cell}}$, $\zeta_1 \approx 1$), $\varphi^{(2\mathrm{R})}/\varphi^{(1\mathrm{R})} \sim 4\times10^6$. Equivalently, to reach the same nonlinear phase the singly-resonant metasurface needs $\sim 10^6$–$10^7\times$ more incident power.

Why the gap is so large. The cascade kernel $\chi^{(2)}G(2\omega_1)\chi^{(2)}$ is enhanced by the SH resonance only. In the 2R case the FM resonance separately enhances the *local intensity* that drives the cascade by Q_1, and the *readout* of the cascaded phase by another Q_1 (field circulates through the cavity for Q_1 round-trips, accumulating phase coherently); hence Q_1^2. Throwing away Q_1 throws away both factors.

Layer 3: what dies, what survives

The following table maps Section 12 of the main note's observables onto the 1R case.

Observable	Scaling with Q_1	Status in 1R
SPM phase per photon	$\propto Q_1^2$	$\sim 10^{-7}$–$10^{-6}\times$ 2R; usable only at high peak power
Cascaded Im $\chi^{(3)}$ (saturable absorber)	$\propto Q_1^2$ (per photon)	survives as a material property; observable contrast tiny
Optical bistability	requires $\varphi_{\mathrm{NL}} \gtrsim 1$ at cavity threshold	*dies*; no FM resonance to be detuned by self-shift
AM modulator (FM-lineshape transduction)	uses FM Lorentzian slope $\propto Q_1$	*dies*; no FM lineshape to slope-detect
Cascade-mediated FWM	$\propto Q_1^2 \cdot Q_2$ for degenerate FM signals	survives as bulk-like FWM, $\propto Q_2$ only
Sum-rule peak $\times$ bandwidth at ω_1	$\propto Q_1\,Q_2$ in 2R	reduces to $\propto Q_2$

Observables that need an FM-resonance feedback loop (bistability, AM modulator, frequency-selective FWM) *die* qualitatively; observables that depend only on the material $\chi^{(3)}_{\mathrm{casc}}$ (SPM phase, Im $\chi^{(3)}$, broadband cascaded FWM) survive but with the $\sim 10^6$–10^7 suppression of (26). The metasurface is then competing with bulk PPLN on raw nonlinear strength, not on the per-photon-phase axis where doubly-resonant cascading shines.

The next section makes the observable-by-observable mapping explicit.

8 Application landscape: observable-by-observable comparison

This section traces each Kerr-driven observable of the main note's §12 through the 1R reduction. The structural finding is uniform: every observable governed by the local cascaded susceptibility $\chi^{(3)}_{\mathrm{casc}}(\omega_1)$ *survives* in 1R with the substitution $\eta_{\mathrm{geom}} \to \eta^{(1\mathrm{R})}_{\mathrm{geom}}$ but loses the Q_1^2 FM-cavity leverage, while every observable that requires *feedback through the FM cavity dynamics* – bistability, all-optical switching, the parametric ceiling, the FM-lineshape-based AM modulator – *dies qualitatively.*

We define the 1R analogue of the main-note ratio $R \equiv |\chi^{(3)}_{\mathrm{casc,(2R)}}|/|\chi^{(3)}_{\mathrm{native}}|$ as

$$R^{(1\mathrm{R})} \;\equiv\; \big|\chi^{(3)}_{\mathrm{casc,(1R)}}\big|/\big|\chi^{(3)}_{\mathrm{native}}\big|\,.$$

For the $LiNbO_3$ design point of §10, $R^{(1\mathrm{R})} \approx 16$ vs. $R \approx 49$ (both linear in their respective η_{geom}, which is the dominant uncertainty); the ratio $R^{(1\mathrm{R})}/R \approx \eta^{(1\mathrm{R})}_{\mathrm{geom}}/\eta_{\mathrm{geom}} \sim 1/3$.

For each item we state the 2R result, give the 1R analogue, quote the *architecture ratio* (cascade vs. native within the same configuration) and the *configuration ratio* (2R vs. 1R at the same observable).

Tier 1: R-scaling observables

1. Self-phase modulation (SPM). 2R: $\varphi_{\rm NL}^{(2R)} = 8\,\alpha_3^{\rm eff}|s_+|^2/\gamma_1^2$ in the fully-radiative limit (main note §12 item 1), with the architecture ratio $(\varphi^{\rm A+B}/\varphi^{\rm A})^{(2R)} = 1+R$. 1R: $\varphi_{\rm NL}^{(1R)}$ from (25), with $\chi_{\rm eff}^{(3)} = \chi_{\rm native}^{(3)} + \chi_{\rm casc,(1R)}^{(3)}$ and the same architecture form $(\varphi^{\rm A+B}/\varphi^{\rm A})^{(1R)} = 1+R^{(1R)}$. Configuration ratio $\varphi^{(2R)}/\varphi^{(1R)} \sim 4\times10^6$ from (26). **Architecture lever survives ($1+R^{(1R)}\sim 17$ vs. $1+R\sim50$); absolute per-photon phase Q_1^2-suppressed and unobservable at CW (see N3 of §10).**

2. Cross-phase modulation (XPM). A separate probe at ω_s (co-resonant with FM in 2R, or merely inside the linear transmission band in 1R) sees an intensity-dependent refractive index driven by the pump. The Boyd index-permutation factor gives $\alpha_3^{\rm XPM} = 2\alpha_3^{\rm SPM}$ in both configurations, and the cascade contributes via pump→SH→DFG-with-signal. The architecture ratio is the same as SPM ($1+R$ in 2R, $1+R^{(1R)}$ in 1R). In 1R the probe is also unenhanced, so the configuration ratio acquires the same Q_1^2 from the pump field plus a further Q_1 from the probe build-up that survives in 2R but not 1R, totalling $\sim Q_1^3$. **Architecture lever survives; absolute Q_1^3-suppressed.**

3. Bistability threshold and 4. all-optical switching. 2R: $|s_+|^2_{\rm bist} \propto \gamma_1^2|\delta_p|/(|\alpha_3^{\rm eff}|\gamma_1^{\rm rad})$ with cusp condition $|\delta_p| > \sqrt{3}\gamma_1/2$, giving architecture ratios $|s_+|^{2,\rm A+B}_{\rm bist}/|s_+|^{2,\rm A}_{\rm bist} = 1/(1+R)$ and $E^{\rm A+B}_{\rm switch}/E^{\rm A}_{\rm switch} = 1/(1+R)$ (main note §12 item 3 – the canonical photonic-computing application). **1R: both die qualitatively.** The cusp condition requires an FM cavity that the pump can nonlinearly detune; without one there is no fixed-point bifurcation in the pump's own resonance. The metasurface transmission at ω_1 acquires a small intensity-dependent phase shift but no hysteresis. This is the most consequential single qualitative loss in 1R: bistability-derived switching is unavailable *at any pump power*, not merely suppressed.

5. Four-wave mixing (FWM). 2R: $\eta_{\rm FWM} \propto |\alpha_3^{\rm eff}|^2|a_p|^4/(\gamma_1/2)^2$, architecture ratio $(1+R)^2$, the most differentiating Tier-1 test of the cascade (main note §12 item 4). Bandwidth limited by $\min(\gamma_1,\gamma_2)$, ∼GHz at telecom Q's. 1R: integrating the SVEA idler equation with $\chi_{\rm casc,(1R)}^{(3)}$ as the four-wave susceptibility,

$$\frac{dA_i}{dz} \;=\; i\,\frac{3\omega_1}{2n_1c}\,\chi_{\rm casc,(1R)}^{(3)}\,A_p^2\,A_s^*\,e^{-i\Delta kz}\,, \tag{27}$$

the idler amplitude after a sub-wavelength metasurface is $A_i \simeq i\,(3\omega_1/(2n_1c))\,\chi_{\rm casc,(1R)}^{(3)}\,A_p^2A_s^*\,L_{\rm meta}$ (the sub-wavelength meta-atom automatically satisfies $\Delta kL_{\rm meta}\ll1$, so $\mathrm{sinc}(\Delta kL_{\rm meta}/2)\to1$ – the bulk PPLN phase-matching constraint does not apply). Conversion efficiency

$$\eta_{\rm FWM}^{(1R)} \;\equiv\; \big|A_i/A_s\big|^2 \;\propto\; |\chi_{\rm casc,(1R)}^{(3)}|^2\,|A_p|^4\,L_{\rm meta}^2\,, \tag{28}$$

inherits the architecture ratio $(1+R^{(1R)})^2$ at fixed local pump intensity. Configuration ratio: 2R wins by $|a_p|^4/|A_p|^4_{\rm 1R} \sim Q_1^2$ (pump field build-up squared) times $1/(\gamma_1/2)^2 \sim Q_1^2$ (FM cavity propagator on the idler) times the R-ratio squared, total $\eta^{(2R)}/\eta^{(1R)} \sim Q_1^4(R/R^{(1R)})^2 \sim 10^{13}$ for $Q_1=10^3$. **Architecture lever survives; absolute efficiency severely Q_1^4-suppressed.** Broadband phase-free operation is the one silver lining: a 1R FWM device tolerates the full γ_2 bandwidth at its full single-pass efficiency, with no FM-bandwidth restriction – useful, perhaps, when bandwidth × efficiency matters more than efficiency alone (cascaded broadband wavelength conversion as a diagnostic of $\chi_{\rm casc,(1R)}^{(3)}$, rather than as a device).

Tier 2: cascade-only capabilities

6. Sign-tunable saturable absorber. 2R: $\mathrm{Im}\,\alpha_3^{\mathrm{eff}}$ peaks at $\Delta\omega = 0$ with Lorentzian profile $|\beta|^2(\gamma_2/2)/[\Delta\omega^2 + (\gamma_2/2)^2]$; $\Delta\omega$-detunable from peak by sweeping pump or SH resonance, suppressing the loss as $(\gamma_2/\Delta\omega)^2$ (main note §12 item 5). 1R: $\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc,(1R)}}(\Delta\omega)$ has the identical Lorentzian structure with peak value (30). Observable single-pass intensity-dependent loss

$$\Delta T/T \;\sim\; \frac{3\,\omega_1\,\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc,(1R)}}\,|E_{\mathrm{loc}}|^2\,L_{\mathrm{meta}}}{n_1\,c}$$

inherits the same Q_1^2 suppression of $|E_{\mathrm{loc}}|^2$ as φ_{NL}. Configuration ratio at fixed $|s_+|^2$ is $\sim Q_1^2(R/R^{(1\mathrm{R})}) \sim 10^6$. **Mechanism and $\Delta\omega$-tunability survive intact; contrast at fixed pump power is Q_1^2-suppressed.**

7. All-optical amplitude modulator via $\Delta\omega$ tuning. 2R: modulating $\Delta\omega(t) = \Delta\omega_0 + \delta\Omega\cos(\Omega_{\mathrm{mod}}t)$ modulates α_3^{eff}, which modulates the FM transmission via the FM-cavity Lorentzian slope; transfer function has a low-pass shoulder ($\Omega_{\mathrm{mod}} \lesssim \gamma_2/2$) plus an AC band-pass peak at $\Omega_{\mathrm{mod}} = |\Delta\omega_0|$ of FWHM γ_2 and peak-to-DC ratio $\sim 4\Delta\omega_0^2/\gamma_2^2$ (main note §12 item 6). 1R: $\delta\chi^{(3)}_{\mathrm{casc,(1R)}}(t)$ via $\Delta\omega(t)$ modulates the single-pass nonlinear phase $\delta\varphi^{(1\mathrm{R})}_{\mathrm{NL}}(t)$, and the AC band-pass structure $\chi^{(3)}_{\mathrm{casc,(1R)}}(\Omega_{\mathrm{mod}})$ peaked at $\Omega_{\mathrm{mod}} = |\Delta\omega_0|$ persists as a property of the cascade sum-rule kernel (§9). But (i) there is no FM Lorentzian to slope-detect against – the FM transmission is broadband – and (ii) $\varphi^{(1\mathrm{R})}_{\mathrm{NL}}$ at attainable CW pump powers is at the μrad level (§10). **The *transduction* from $\Delta\omega(t)$ to readable FM amplitude modulation dies as a practical capability.** The cascade still carries the AC band-pass signature, but it is not coupled into an observable signal channel.

8. Passive optical limiter (parametric ceiling). 2R: hard cap $|a_1|^2_{\mathrm{cap}} = (\Delta\omega/\beta)^2$ above which the FM trivial fixed point destabilises (cascaded OPO threshold); the FM clamps regardless of further pump increase, with excess input dissipated through the cascaded TPA channel (main note §12 item 7). 1R: **the parametric-instability mechanism dies**. The FM is not a dynamical mode in 1R; its amplitude inside the meta-atom is fixed externally by the incident pump and the linear scattering response of the structure, not by a fixed point of a TCMT equation. There is no trivial fixed point to lose stability, no OPO-like clamping. A weaker form of intensity clamping does survive – the cascaded TPA of item 6 gives $\Delta T/T \propto \mathrm{Im}\,\chi^{(3)}_{\mathrm{casc,(1R)}}\,|E_{\mathrm{loc}}|^2$, which at extreme pump intensities approaches unity – but this is the saturable absorber re-framed as a power limiter, not the hard dispersive ceiling of the 2R parametric instability.

9. $\Delta\omega$-readout sensor. 2R: at $\Delta\omega = 0$, $|d\alpha_3^{\mathrm{casc}}/d\Delta\omega| = |\beta|^2/(\gamma_2/2)^2 \propto Q_2^2$, giving sensor sensitivity $|d\varphi_{\mathrm{NL}}/d\Delta\omega| = 32|s_+|^2|\beta|^2/(\gamma_1^2\gamma_2^2) \propto Q_1^2Q_2^2$ (main note §12 item 8); shot-noise-limited FoM slope$/\sqrt{\gamma_3^{\mathrm{eff}}} \propto Q_2^{3/2}$. 1R: differentiating $\chi^{(3)}_{\mathrm{casc,(1R)}} = -2\varepsilon_0\omega_1|\widetilde{M}|^2/[3N_2(\Delta\omega + i\gamma_2/2)V_{\mathrm{cell}}]$,

$$\left|\frac{d\chi^{(3)}_{\mathrm{casc,(1R)}}}{d\Delta\omega}\right|_{\Delta\omega=0} = \frac{2\,\varepsilon_0\,\omega_1\,|\widetilde{M}|^2}{3\,N_2\,(\gamma_2/2)^2\,V_{\mathrm{cell}}} = \frac{2\,Q_2^2\,|\chi^{(2)}|^2\,\eta^{(1\mathrm{R})}_{\mathrm{geom}}}{3\,\omega_1\,n_2^2} \;\propto\; Q_2^2\,, \tag{29}$$

the same Q_2^2 scaling of the *material slope* as 2R. The observable single-pass phase sensitivity inherits this: $|d\varphi^{(1\mathrm{R})}_{\mathrm{NL}}/d\Delta\omega| \propto Q_2^2$ (no Q_1^2). The shot-noise FoM scales as $Q_2^{3/2}$ as in 2R, since both slope and $\mathrm{Im}\,\chi^{(3)}$ have the same Q_2 dependence and cancel one power. **Mechanism and Q_2 scaling survive intact; absolute sensitivity per unit perturbation is Q_1^2-suppressed, matching the φ_{NL} suppression.**

Compact comparison table

Observable	2R archit. ratio	1R archit. ratio	Config. ratio $X^{(2R)}/X^{(1R)}$
SPM phase	$1+R$	$1+R^{(1R)}$	$\sim Q_1^2 \cdot (R/R^{(1R)}) \sim 10^6$
XPM phase (probe)	$1+R$	$1+R^{(1R)}$	$\sim Q_1^3 \cdot (R/R^{(1R)}) \sim 10^9$
Bistability threshold	$1/(1+R)$	— (dies)	—
All-optical switching energy	$1/(1+R)$	— (dies)	—
FWM efficiency	$(1+R)^2$	$(1+R^{(1R)})^2$	$\sim Q_1^4 \cdot (R/R^{(1R)})^2 \sim 10^{13}$
Saturable absorber ($\Delta T/T$)	cascade-only	cascade-only	contrast $\sim Q_1^2 \sim 10^6$
AM modulator (readout)	cascade-only	— (dies as readout)	—
Optical limiter (parametric)	cascade-only	— (dies)	—
Sensor slope $d\chi^{(3)}/d\Delta\omega$	$\propto Q_2^2$	$\propto Q_2^2$	material-slope: ~ 1
Sensor phase sensitivity	$\propto Q_1^2 Q_2^2$	$\propto Q_2^2$	$\sim Q_1^2 \sim 10^6$

Pattern. Items that *die qualitatively* in 1R are precisely those that require the FM cavity to be *the nonlinear oscillator*: bistability turns the FM cavity into a self-referenced nonlinear element via the detuning $\alpha_3^{\text{eff}}|a_1|^2$; the parametric ceiling is the instability of the FM cavity's trivial fixed point under SH back-action; the AM modulator transduces $\Delta\omega(t)$ through the FM cavity's Lorentzian slope into a transmission modulation. Removing the FM cavity removes all three.

Items that *survive but suppress* are those where $\chi^{(3)}_{\text{casc}}(\omega_1)$ is the material input to a single-pass or small-signal observable – SPM, XPM, cascaded TPA, FWM, sensor material-slope. Each is suppressed by the Q_1-related buildup/readout factor specific to that observable (quadratic for SPM/saturable absorber/sensor, cubic for XPM, quartic for FWM). None recover an observable signal at CW pump powers below material damage.

9 Bandwidth, sum rule, and sign-tunability in 1R

Three structural features of the doubly-resonant case generalise.

Sign tunability. The denominator $(\Delta\omega + i\gamma_2/2)$ in (15) is identical to the doubly-resonant case, so $\mathrm{Re}\,\chi^{(3)}_{\text{casc},(1R)} \propto -\Delta\omega/(\Delta\omega^2+\gamma_2^2/4)$ flips sign at $\Delta\omega = 0$. Sweeping the pump wavelength through $\Delta\omega = 0$ tunes between self-focusing ($\Delta\omega < 0$) and self-defocusing ($\Delta\omega > 0$). In the 1R case this sweep is bounded only by how broadband the incident source is and by the dispersion of E_{loc} across the sweep; in particular, there is no FM resonance to fall off. This is *cleaner* than the 2R case, where the pump-tuning range is bounded by γ_1.

Two-photon-like loss. $\mathrm{Im}\,\chi^{(3)}_{\text{casc},(1R)} \propto (\gamma_2/2)/(\Delta\omega^2 + \gamma_2^2/4)$ peaks at $\Delta\omega = 0$ with the same Lorentzian profile as in 2R. Setting $\Delta\omega = 0$ in (15) and using $|\widetilde{M}|^2/(N_2 V_{\text{cell}}) = \eta^{(1R)}_{\text{geom}}|\chi^{(2)}|^2/(\varepsilon_0 n_2^2)$ together with $\omega_1/\gamma_2 = Q_2/2$,

$$\mathrm{Im}\,\chi^{(3)}_{\text{casc},(1R)}\big|_{\Delta\omega=0} = \frac{2\,Q_2\,|\chi^{(2)}|^2\,\eta^{(1R)}_{\text{geom}}}{3\,n_2^2}, \tag{30}$$

in $\mathrm{m}^2/\mathrm{V}^2$, of the same magnitude as the doubly-resonant value with the η-substitution. Whether

this loss is observable as a saturable absorption at the metasurface level depends on $\varphi_{\mathrm{NL}}^{(1\mathrm{R})}$ being measurable – which, per Section 7, requires extreme pump intensities.

Sum rule (modified). The doubly-resonant sum rule $|\chi^{(3)}_{\mathrm{casc}}|_{\max}\times\Delta f \lesssim \omega_1|\chi^{(2)}|^2\eta_{\mathrm{geom}}/(3\pi n_2^2)$ generalises with $\eta_{\mathrm{geom}} \to \eta^{(1\mathrm{R})}_{\mathrm{geom}}$; the Q_2-independent peak×bandwidth product survives because it is an SH-resonance property only. The frequency-domain bandwidth at the design point is unchanged: $\Delta f_{\mathrm{usable}} \sim \gamma_2/(2\pi) \simeq 97$ GHz at $Q_2 = 4\times10^3$.

10 Worked numerical estimate

Use the $LiNbO_3$ design point: $\lambda_1 = 1.55\,\mu$m so $\omega_1 = 2\pi\cdot 193.5\,$THz, $|\chi^{(2)}| = 5\times10^{-11}\,$m/V, $n_1 \simeq 2.21$, $n_2 \simeq 2.26$. SH resonance $Q_2 = 4\times10^3$ giving $\gamma_2/(2\pi) \simeq 97\,$GHz; conservative detuning $\Delta\omega = 2\gamma_2$; geometric overlap $\eta^{(1\mathrm{R})}_{\mathrm{geom}} = 0.1$ (the 1R working number from Section 6). Meta-atom thickness $L_{\mathrm{meta}} = 400\,$nm; $T_{\mathrm{geom}} = 1$; unit cell area $A_{\mathrm{cell}} = (500\,\mathrm{nm})^2$.

Effective $\chi^{(3)}_{\mathbf{casc},(1\mathrm{R})}$. From (18),

$$|\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}| \;\sim\; \frac{\omega_1\,|\chi^{(2)}|^2\,\eta^{(1\mathrm{R})}_{\mathrm{geom}}}{3\,n_2^2\,\gamma_2} \;=\; \frac{Q_2}{6\,n_2^2}\,|\chi^{(2)}|^2\,\eta^{(1\mathrm{R})}_{\mathrm{geom}} \;\simeq\; 3.3\times10^{-20}\,\mathrm{m^2/V^2}\,, \tag{31}$$

a factor of ~ 16 above bulk $LiNbO_3$ Kerr, factor of ~ 3 below the 2R prediction.

Per-photon nonlinear phase. From (25) with $|s_+|^2 = P_{\mathrm{in}} = 1\,$mW per unit cell (an unrealistically optimistic per-meta-atom power, used here only for benchmarking):

$$\varphi_{\mathrm{NL}}^{(1\mathrm{R})} \;\sim\; \frac{3\,\omega_1\,L_{\mathrm{meta}}}{4\,\varepsilon_0\,c^2\,n_1^3\,A_{\mathrm{cell}}}\,\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}\,P_{\mathrm{in}} \;\sim\; 5\times10^{-9}\,\mathrm{rad}. \tag{32}$$

(This value is unchanged by the October 2026 correction: the factor 4 in $\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}$ is compensated by the Boyd-convention $|E_{\mathrm{loc}}|^2$, which is 4× smaller than the value used before.) Compare with the doubly-resonant case at the same per-cell power and $Q_1 = 10^3$, from $\varphi^{(2\mathrm{R})} = 8|\beta|^2|s_+|^2/(\gamma_1^2\Delta\omega)$ with $\eta_{\mathrm{geom}} = 0.3$, $V_{\mathrm{eff}} = V_{\mathrm{cell}}$ and $\zeta_1 = 1$: $\varphi^{(2\mathrm{R})}_{\mathrm{NL}} \sim 2\times10^{-2}\,$rad. Ratio $\sim 4\times10^6$, consistent with (26) including its $\mathcal{O}(1)$ prefactor.

Damage threshold check. To reach 0.1 rad in 1R requires $P_{\mathrm{in}} \sim 0.1/(5\times10^{-9}) \times 1\,\mathrm{mW} \simeq 20\,$kW per unit cell. At unit-cell area $A_{\mathrm{cell}} = (500\,\mathrm{nm})^2 = 2.5\times10^{-13}\,\mathrm{m^2}$ this is an intensity $\sim 8\times10^{12}\,\mathrm{W/cm^2}$, three to four orders of magnitude above the ns-pulse damage threshold for $LiNbO_3$ ($\sim 1-10\,\mathrm{GW/cm^2}$). Even at damage-threshold intensity ($\sim 25\,$W per unit cell for $10\,\mathrm{GW/cm^2}$), the accumulated nonlinear phase is only $\varphi^{(1\mathrm{R})}_{\mathrm{NL}} \sim 10^{-4}\,$rad – borderline interferometrically detectable but well below operating regimes for any Kerr-based device. The single-pass SH-only metasurface therefore does not deliver an observable CW or quasi-CW Kerr nonlinearity at ω_1; quoting it as "high peak power" overstates the case.

Bandwidth-integrated nonlinear strength $|\chi^{(3)}|\cdot\Delta f$. At $\Delta f \sim \gamma_2/(2\pi) \simeq 97\,$GHz,

$$|\chi^{(3)}_{\mathrm{casc},(1\mathrm{R})}|\cdot\Delta f \;\sim\; 3.3\times10^{-20}\cdot 9.7\times10^{10} \;\sim\; 3.2\times10^{-9}\,\mathrm{m^2\,Hz/V^2}\,. \tag{33}$$

Bulk PPLN at conservative parameters gives $|\chi^{(3)}_{\mathrm{casc,bulk}}|\cdot\Delta f \sim 2\times10^{-8}\,\mathrm{m^2\,Hz/V^2}$ (main note §10, after the 2026-05 $1/n_2$ correction; cm path length). The 1R metasurface is a factor ~ 6

below bulk PPLN in this benchmark – a gap that the doubly-resonant case can only *partially* close (factor ~ 3 from η_{geom}). The metasurface's value proposition with one resonance only is therefore chip-scale form factor and CW operation at modest threshold, not raw $|\chi^{(3)}|\cdot\Delta f$ competitiveness.

11 Summary

Boxed principal result.

$$\boxed{\chi^{(3)}_{\mathrm{casc,(1R)}}(\omega_1) \;=\; -\,\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2}\,\frac{|\widetilde{M}|^2}{(\Delta\omega + i\,\gamma_2/2)\,V_{\mathrm{cell}}} \;\simeq\; -\,\frac{2\,\omega_1\,|\chi^{(2)}|^2\,\eta^{(1\mathrm{R})}_{\mathrm{geom}}}{3\,n_2^2\,\Delta\omega}\,,}$$

with the singly-resonant SHG overlap and geometric factor

$$\widetilde{M} \;\equiv \int_{V_{\mathrm{NL}}} \chi^{(2)}_{ijk}(\mathbf{r})\, f^{*}_{2,i}(\mathbf{r})\,\hat{e}_j\,\hat{e}_k\; d^3r, \qquad \eta^{(1\mathrm{R})}_{\mathrm{geom}} \;\equiv\; \frac{|\widetilde{M}|^2\,\varepsilon_0\,n_2^2}{N_2\,V_{\mathrm{cell}}\,|\chi^{(2)}|^2} \;\sim\; 0.05{-}0.2\,,$$

with $N_2 = \frac{1}{2}$ the projection norm of the energy-normalised SH mode (the prefactor 2/3 replaces the 1/6 of earlier versions, which was 4× too small; see the main note, §5).

Comparison table.

Quantity	2R (doubly-resonant)	1R (SH-only)
SHG overlap	$M = \int \chi^{(2)} f_2^* \mathbf{f}_1 \mathbf{f}_1$	$\widetilde{M} = \int \chi^{(2)} f_2^* \hat{\mathbf{e}}\hat{\mathbf{e}}$
Normalisation volume	$\int \lvert\mathbf{f}_1\rvert^4\, d^3r$	V_{cell}
Geometric overlap η_{geom}	$\widetilde{\eta}^2/\zeta_1 \sim 0.1{-}0.5$	$\lvert\widetilde{M}\rvert^2\varepsilon_0 n_2^2/(N_2 V_{\mathrm{cell}}\lvert\chi^{(2)}\rvert^2) \sim 0.05{-}0.2$
Structural form of $\chi^{(3)}_{\mathrm{casc}}$	identical: $-2\omega_1\lvert\chi^{(2)}\rvert^2\eta_{\mathrm{geom}}/(3n_2^2\Delta\omega)$	
Sign tunability by $\Delta\omega$	yes	yes (broader tuning window)
Two-photon-like loss	$2Q_2\lvert\chi^{(2)}\rvert^2\eta_{\mathrm{geom}}/(3n_2^2)$ at $\Delta\omega = 0$	same prefactor with $\eta \to \eta^{(1\mathrm{R})}$
Sum rule $\lvert\chi^{(3)}_{\mathrm{casc}}\rvert\cdot\Delta f$	$\sim \omega_1\lvert\chi^{(2)}\rvert^2\eta_{\mathrm{geom}}/(3\pi n_2^2)$	same with $\eta \to \eta^{(1\mathrm{R})}$
Per-photon φ_{NL}	$\propto Q_1^2 \cdot Q_2$	$\propto Q_2$ only
Bistability, AM modulator	yes	no
$\lvert\chi^{(3)}_{\mathrm{casc}}\rvert\cdot\Delta f$ vs. bulk PPLN (2×10^{-8}, §10)	factor ~ 0.5 (PPLN wins by $\sim 2\times$)	factor ~ 0.16 (PPLN wins by $\sim 6\times$)

Take-away. The cascaded $\chi^{(3)}$ *mechanism* is essentially intact when the FM resonance is removed: $G(2\omega_1)$ alone carries the resonant kernel and the boxed structural form is unchanged. What collapses is the *leverage* of that material response on the observable per-photon phase, because Q_1 enters twice in the doubly-resonant phase (once in field build-up, once in readout) and is absent in the SH-only case. Designs that need only $\chi^{(3)}_{\mathrm{casc}}$ as a material parameter (e.g. broadband cascaded FWM, certain pulsed-saturable-absorber schemes) tolerate the SH-only configuration; designs that need feedback through an FM resonance (bistability, AM modulator, resonant phase matching for SPM-based switching at CW thresholds) do not.

A Quick prefactor check from the doubly-resonant limit

A useful sanity check is to recover the doubly-resonant boxed result by "adding back" an FM resonance. The substitution is not literal – M and $\widetilde{M}$ have different units, since $\mathbf{f}_1$ carries

$\mathrm{V}/(\mathrm{m}\sqrt{\mathrm{J}})$ while $\hat{\boldsymbol{e}}$ is dimensionless – but a structural correspondence holds at the level of the SHG drive. Read (5) as the projection of $P^{(2)}(\mathbf{r}, 2\omega_1) = \varepsilon_0 \chi^{(2)}\, E(\mathbf{r}, \omega_1)\, E(\mathbf{r}, \omega_1)$ onto $\mathbf{f}_2^*$, with the *local FM field* substituted for $E(\mathbf{r}, \omega_1)$. In the 1R case that field is $E_{\mathrm{loc}}\, \hat{\boldsymbol{e}}$ (uniform c-number, magnitude scaled out as E_{loc}^2 outside the integral), giving $\hat{\beta} = (\varepsilon_0 \omega_1 / N_2)\, E_{\mathrm{loc}}^2 \widetilde{M}$; in the 2R case it is $a_1 \mathbf{f}_1(\mathbf{r})$ (modal, amplitude scaled out as a_1^2 outside), giving $\beta a_1^2 = (\varepsilon_0 \omega_1 / N_2)\, a_1^2 M$.

Identifying $E_{\mathrm{loc}}^2 \leftrightarrow a_1^2$ and $\hat{\boldsymbol{e}}\hat{\boldsymbol{e}} \leftrightarrow \mathbf{f}_1(\mathbf{r})\mathbf{f}_1(\mathbf{r})/[\mathbf{f}_1\text{-normalisation}]$ at the integrand level, and likewise replacing the unit-cell volume V_{cell} in the cascade-back-projection denominator of (15) by the modal participation volume $\int |\mathbf{f}_1|^4\, d^3r$, the structural form

$$\chi^{(3)}_{\mathrm{casc,(1R)}} \;\longleftrightarrow\; -\,\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2}\,\frac{|M|^2}{(\Delta\omega + i\gamma_2/2)\int |\mathbf{f}_1|^4\, d^3r} \;=\; \chi^{(3)}_{\mathrm{casc,(2R)}}\,,$$

matches the main note Eq. (chi3casc). The prefactor $2\varepsilon_0\omega_1/3N_2$ and the denominator structure $\Delta\omega + i\gamma_2/2$ are common to the two cases; what transforms together is the choice of *local-FM-field parameterisation* (uniform-pump c-number vs. modal amplitude) and the matching overlap-volume normalisation (unit cell volume vs. modal participation volume).

References

[1] K. Koshelev, S. Lepeshov, M. Liu, A. Bogdanov, and Y. Kivshar, *Asymmetric Metasurfaces with High-Q Resonances Governed by Bound States in the Continuum*, Phys. Rev. Lett. **121**, 193903 (2018).